E-VFU: A Multi-Attack Verification Framework for Vertical Federated Unlearning

CSE-472 Project

ID: 2005030 ID: 2005034

Outline

1. Outline
2. Background & Problem
3. Our Approach: E-VFU
4. Experimental Setup
5. Results — Figures
6. Conclusions

What is Vertical Federated Learning (VFL)?

- In **horizontal** FL: each party has *different samples* (rows) but same feature set.
- In **vertical** FL: each party has *different features* (columns) for the **same entities** (e.g., same users, same patients).

Example

Party A (hospital) has vitals and diagnoses; Party B (lab) has blood tests. Same patient ID, different columns. They train one model by sharing only **embeddings** or gradients, not raw data.

Used in healthcare, finance, and cross-institutional analytics where data cannot be centralized.

Why Unlearning in VFL?

1. **Right to erasure** (GDPR, etc.): A user can request that their data be removed not only from databases but also from any *model trained on it*.
2. In VFL, "removing data" can mean:
   - removing a **client** (one party leaves),
   - removing a **feature** (one column must no longer influence the model),
   - or removing specific **samples** (rows).
3. Simply retraining from scratch is expensive; we need **unlearning** methods that update the model without full retraining.

Challenge: How do we know unlearning actually worked? We need **verification**.

Base Work: Varshney et al. (PETS 2025)

- First **feature-level** unlearning methods for VFL: **VFU-KD** (knowledge distillation) and **VFU-GA** (gradient ascent).
- Idea: active party retrains (e.g., student mimics teacher on *retain set only*, or gradient ascent on target samples); passive parties may freeze encoders.
- They evaluate unlearning using **output-logit membership inference**: can an attacker tell if a sample was in the training set by looking at model predictions?

Limitation

Output-only MIA is **weak**. Recent work (EMNLP 2025) shows that stronger, unlearning-specific MIA can reach $\sim$99% accuracy even when standard metrics suggest "successful" unlearning. So we need **stronger verification**.

Problem: Verification Gaps

1. **Weak MIA**: Relying only on output logits misses leakage in gradients and internal representations.
2. **No feature entanglement**: If features are correlated, removing one may still leave traces in other parties' encodings; no analysis in prior work.
3. **No communication cost**: Overhead of unlearning in secure aggregation (SMPC/SecAgg) is unstudied.
4. **Single-shot only**: No evaluation of repeated unlearning requests (iterative setting).

E-VFU focuses on (1): we add **stronger adversarial verification** so we can rigorously check whether unlearning really removed the target data.

E-VFU in One Slide

What is E-VFU?

Enhanced Vertical Federated Unlearning: a **verification framework** that audits Varshney et al.'s VFU-KD/VFU-GA using four complementary checks. We do *not* propose a new unlearning algorithm; we propose *better ways to test* if unlearning worked.

Phase 0 (base)

We use the same base as the paper: VFL model (bottom encoders per party + top fusion), training, and unlearning (e.g., gradient ascent on target samples). We also keep **output-based MIA** as a **baseline**.

Phase 1 (robust verification)

We run **four** verification attacks. If all pass $\Rightarrow$ VFU unlearning is robust. If any fails $\Rightarrow$ we have evidence of remaining leakage.

Phase 1 — The Four Verification Components

1. **Output MIA (baseline)**: Attacker uses prediction confidence/loss to guess membership. *Target:* AUC ≈ 0.5 on unlearned samples (cannot distinguish from non-members).
2. **Gradient-based MIA**: Attacker uses *gradient* norms and spectral features to guess membership. Stronger than output-only. *Target:* AUC ≈ 0.5; we also report gradient change $\|g_{\text{before}}\| - \|g_{\text{after}}\|$.
3. **FUMA**: Forensic Unlearning Membership Attack — use *canary* samples and prediction divergence Δ_{pred} to detect unlearning artifacts. *Target:* detection rate $< 5\%$.
4. **Shadow model attack**: Train shadow models on retain-only vs full data; meta-classifier predicts which dataset a model came from. *Target:* unlearned model classified as retain-only (P(full data) < 0.5).

Why These Four?

- **Output MIA**: Standard baseline (Varshney et al.); easy to compare.
- **Gradient MIA**: Gradients reveal more than logits (Geiping et al.); if unlearning is real, gradient response to unlearned samples should change.
- **FUMA**: Unlearning can leave *behavioral* traces (e.g., sensitivity to canaries); FUMA checks that such traces are gone.
- **Shadow attack**: Checks the *whole model*, not per-sample membership — is the unlearned model statistically indistinguishable from a model that never saw the target data?

Together they cover: *outputs*, *gradients*, *behavior on probes*, and *model-level fingerprint*.

Phase 1 — Success Criteria (Summary Table)

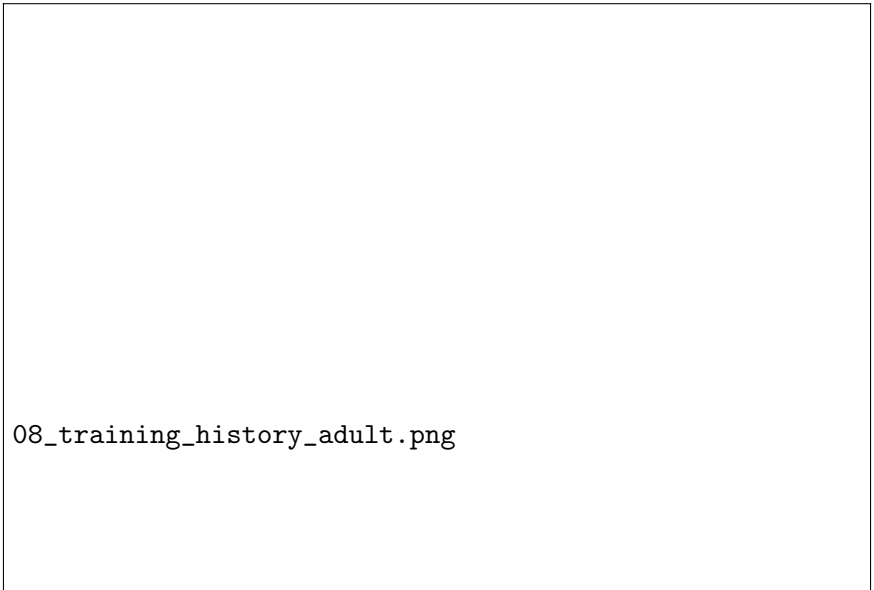
Component	Metric	Interpretation
Output MIA	AUC on unlearned samples	$\approx 0.5 \Rightarrow$ attacker cannot tell member from non-member
Gradient MIA	AUC; gradient change	≈ 0.5 and large gradient change $\Rightarrow$ gradient signature of target data removed
	Unlearning effectiveness	$\geq 50\% \Rightarrow$ unlearned samples look like non-members
FUMA	Detection rate	$< 5\% \Rightarrow$ forensic traces minimal
Shadow model	P(full data) for unlearned model	$< 0.5 \Rightarrow$ model looks like retain-only

In the following slides we show **experimental results** for each component and then the combined dashboard.

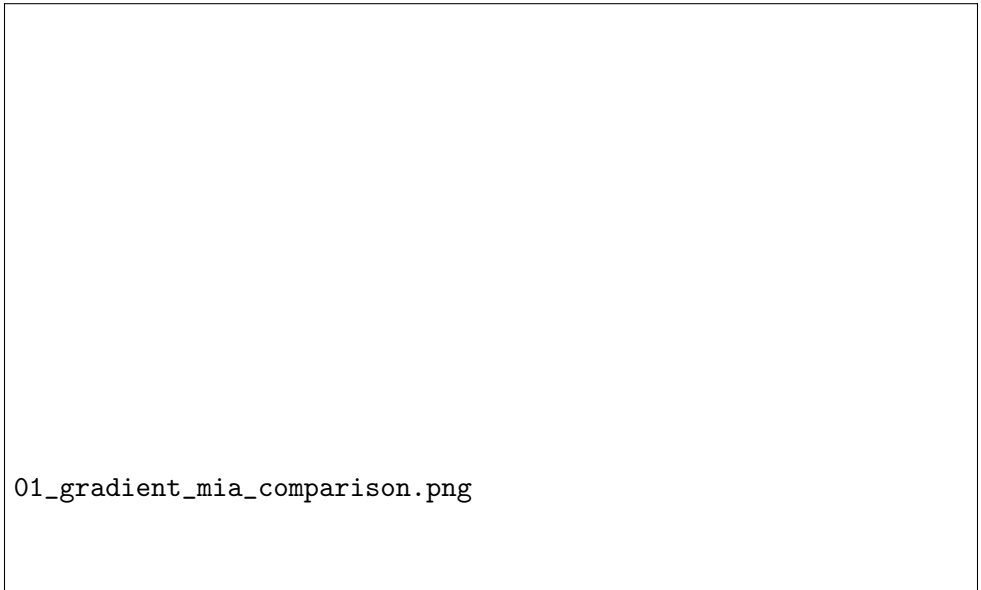
Experimental Setup

- **Datasets**: UCI Adult (income prediction, $\sim$32K samples), German Credit (credit risk, 1K); optional MIMIC-III (healthcare, 46K).
- **VFL**: Data split by **features** across 2 (or 3) parties; one active party holds labels.
- **Model**: Bottom MLPs per party $\rightarrow$ embeddings $\rightarrow$ top fusion MLP $\rightarrow$ binary prediction.
- **Unlearning**: We use **VFU-GA** (gradient ascent on target samples for a fixed number of epochs, e.g., 10). A fraction (e.g., 5%) of training samples are selected as "target" to unlearn.
- **Verification**: After unlearning, we run all four Phase 1 attacks and compare metrics to the criteria above.

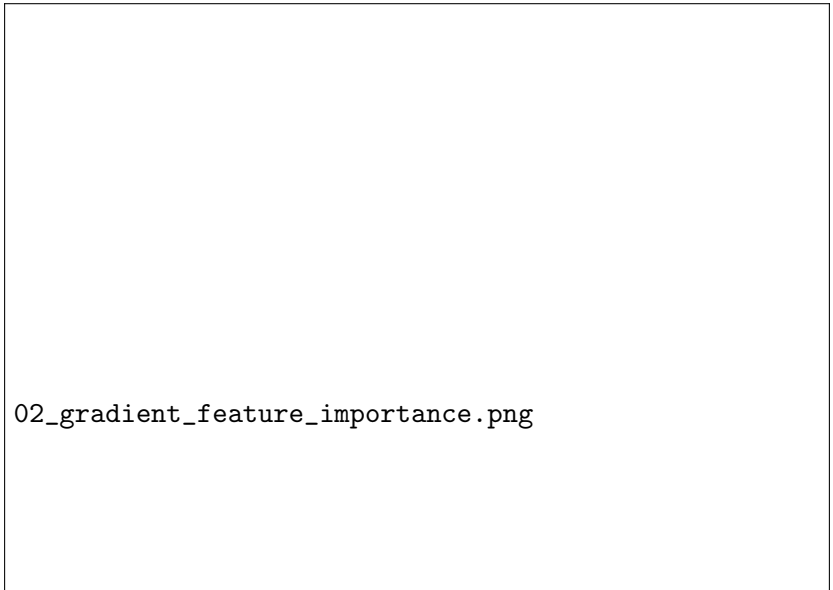
Result 1: Training Curves (Before Unlearning)



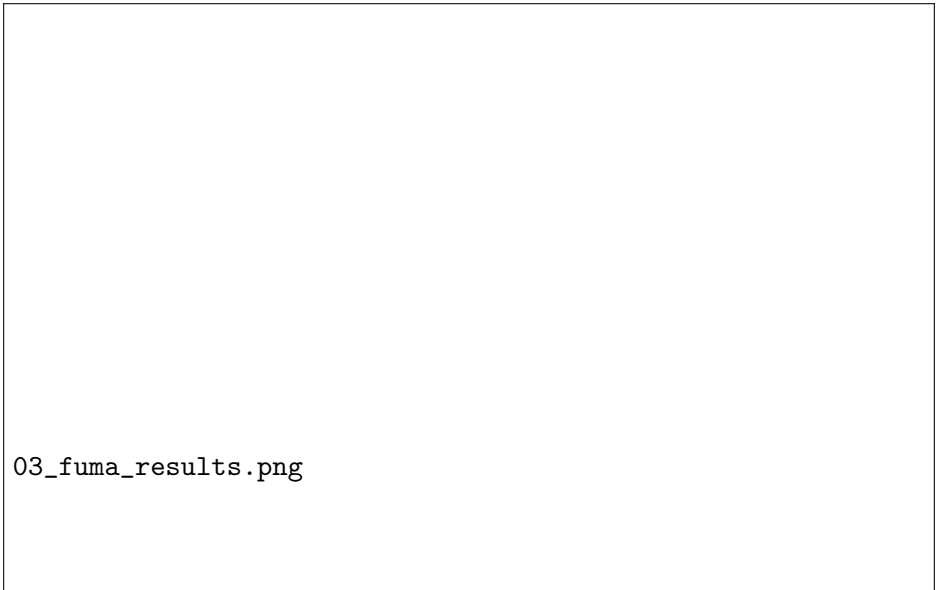
Result 2: Gradient MIA vs Output MIA



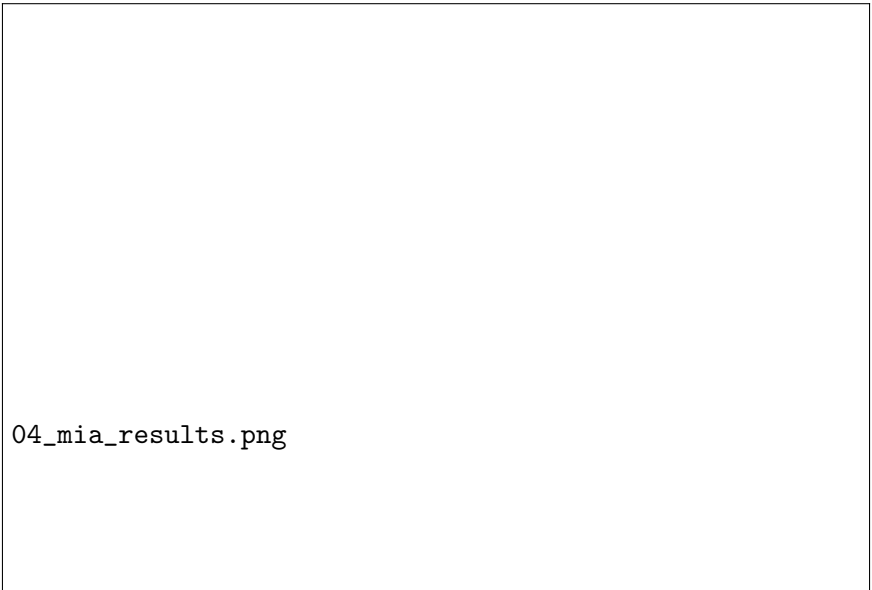
Result 3: Which Gradient Features Matter for MIA?



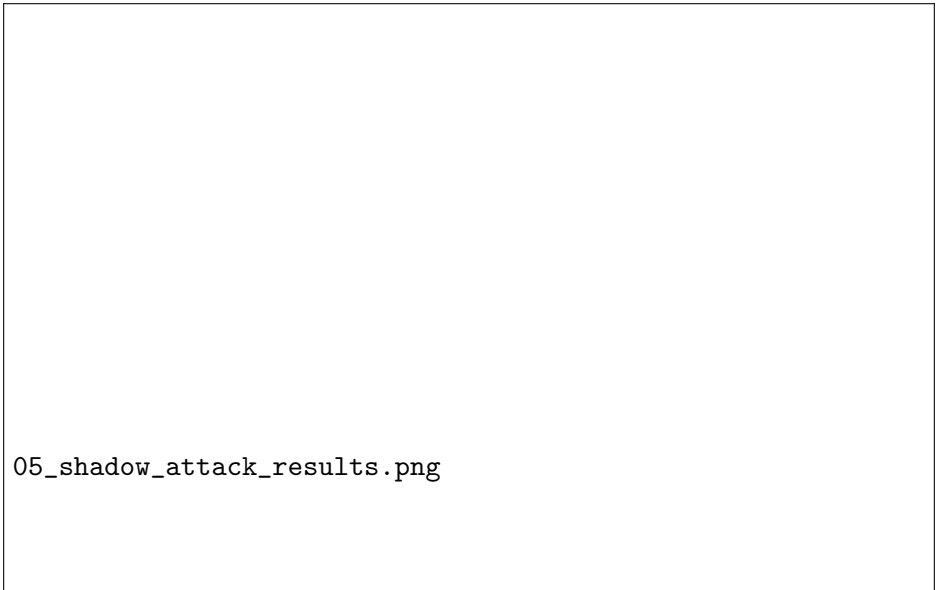
Result 4: FUMA — Forensic Unlearning Membership Attack



Result 5: Baseline (Output-Logit) MIA



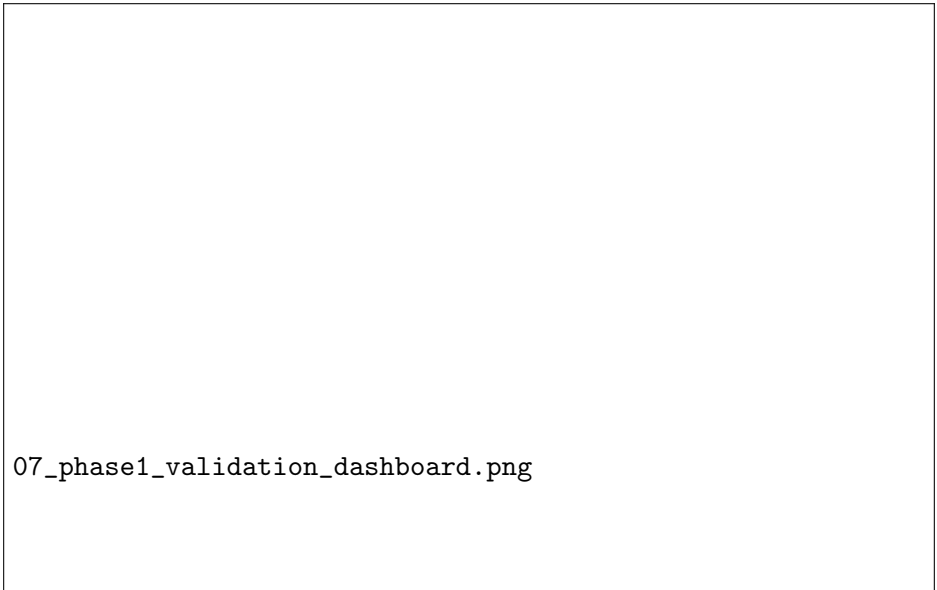
Result 6: Shadow Model Attack



Result 7: Comprehensive Verification (All Metrics + Pass/Fail)



Result 8: Phase 1 Validation Dashboard



What We Did

- We implemented **Phase 0** (VFL model, training, VFU-GA unlearning) and **Phase 1** (four verification attacks) as in the E-VFU proposal.
- We compared **output-based MIA** (baseline) with **gradient-based MIA**, and added **FUMA** and **shadow model** verification.
- Each attack has a clear **success criterion**; we report whether unlearning passes or fails each check.
- All plots are saved separately in `presentation_figures/` (numbered 01–08) for easy inclusion in reports and slides.

Interpretation & Takeaway

If verification passes (all four checks)

E-VFU provides evidence that Varshney et al.'s VFU unlearning **removes** the target data's influence from the model, not only in terms of output logits but also in gradients, forensic behavior, and model-level statistics. The baseline (output MIA) alone would be insufficient to conclude this.

If verification fails (one or more checks)

E-VFU **reveals limitations**: some trace of the target data remains (e.g., in gradients or in the shadow classifier's view). That motivates stronger unlearning methods or more epochs, and shows the value of multi-attack verification.

Bottom line: Stronger verification leads to more reliable conclusions about whether unlearning actually worked.

Deliverables & References

- **Jupyter notebook**: Phase 0 (train, unlearn) + Phase 1 (all four attacks, visualizations).
- **presentation_figures/**: 8 PNGs (01–08) for slides and report.
- **This presentation**: Elaborated explanation and one slide per main result.

Key references: Varshney et al. (PETS 2025) — VFU-KD/VFU-GA; Geiping et al. (2020) — gradient inversion; Shokri et al. — shadow models; EMNLP 2025 — unlearning-specific MIA.

Thank you

Questions?